

ROBERT ANADU

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SUMMARY

Aerospace Engineering graduate with strong foundations in control systems, structural analysis, and system-level modeling. Experienced in designing and validating multidisciplinary engineering systems from first-principles modeling through simulation and performance evaluation. Demonstrates systems thinking, analytical rigor, and the ability to translate theoretical models into structured, testable engineering solutions.

TECHNICAL SKILLS

Control & Systems Engineering: Feedback Control, PID Design & Tuning, System Linearization, Trajectory Planning, Stability & Transient Analysis, Sensor Integration

Modeling & Simulation: MATLAB, Simulink, First-Principles Modeling, System Identification (**tfest**), Performance Validation

Mechanical & Structural: Load Path Analysis, Parametric CAD Modeling, Structural Layout Design, Engineering Documentation

Programming: Python, C/C++, Git

Engineering Tools: SolidWorks, CATIA, ANSYS, LaTeX

PROJECTS

Autonomous 3-DOF Helicopter Control & Trajectory Tracking

Apr 2024

Toronto Metropolitan University

Toronto, ON

- Designed and implemented quintic (5th-order) trajectory generation under actuator and stability constraints to produce dynamically feasible command profiles.
- Developed and tuned multi-axis closed-loop PID controllers within a dual-loop architecture to achieve stable tracking performance.
- Contributed to nonlinear dynamic modeling and equilibrium linearization to derive transfer functions for elevation, pitch, and travel axes.
- Applied system identification techniques to refine plant models and improve controller tuning accuracy.
- Integrated filtering and disturbance rejection methods, including notch filtering and complementary filtering, to enhance robustness under vibration and sensor bias conditions.
- Executed structured simulation-based validation to assess subsystem interaction, closed-loop stability, and tracking performance.

Aircraft Structural Redesign: Wing Box & Landing Gear

Apr 2023

Toronto Metropolitan University

Toronto, ON

- Developed conceptual wing-box and landing gear structures using parametric CAD and first-order load calculations to establish realistic structural load paths.
- Produced section views and structured documentation to communicate structural assumptions, interfaces, and manufacturability considerations.

EXPERIENCE

Facilities Technical Support Technician

Dec 2024 – Present

BGIS

Toronto, ON

- Interpreted engineering drawings and technical specifications to execute precision installations within spatial and safety constraints.
- Performed structured diagnostics and root-cause analysis to restore system functionality in time-critical environments.
- Documented layouts and procedures to support verification, compliance, and continuous improvement initiatives.
- Coordinated across multiple trades while maintaining quality and operational standards.

EDUCATION

Toronto Metropolitan University (formerly Ryerson University)

Sept 2020 – Jun 2024

Bachelor of Engineering in Aerospace Engineering

Toronto, ON

- Relevant Coursework: Control Systems, Flight Dynamics, Avionics Design, Structural Analysis, Fluid Mechanics, Space Robotics

CERTIFICATIONS

Google Cybersecurity Professional Certificate

2025

Coursera

Online

- Completed hands-on labs in secure system architecture and threat modeling applicable to safety-critical engineering environments.